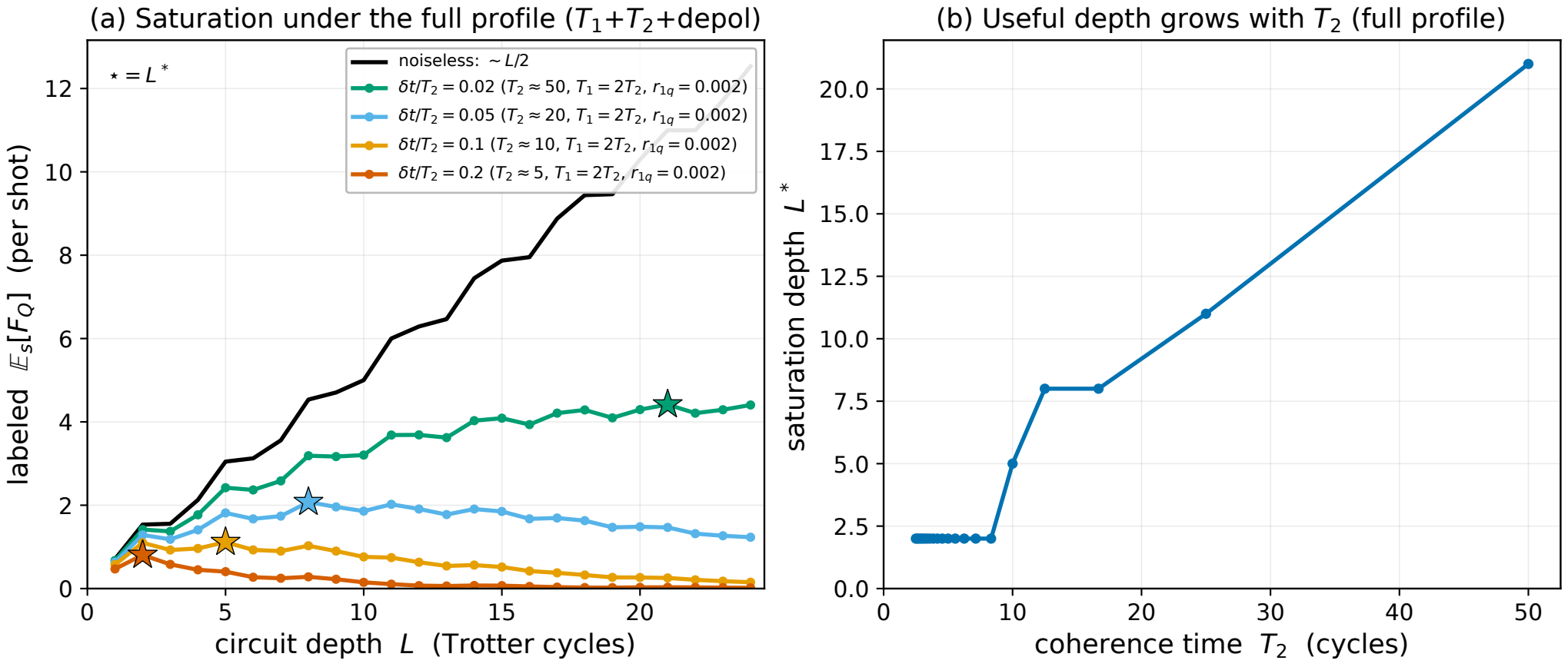
(a) Saturation under the full profile ($T_1$+$T_2$+depol)
$\star = L^*$
noiseless: $\sim L/2$
$\delta t/T_2 = 0.02$ ($T_2 \approx 50$, $T_1 = 2T_2$, $r_{1q} = 0.002$)
$\delta t/T_2 = 0.05$ ($T_2 \approx 20$, $T_1 = 2T_2$, $r_{1q} = 0.002$)
$\delta t/T_2 = 0.1$ ($T_2 \approx 10$, $T_1 = 2T_2$, $r_{1q} = 0.002$)
$\delta t/T_2 = 0.2$ ($T_2 \approx 5$, $T_1 = 2T_2$, $r_{1q} = 0.002$)
labeled $\mathbb{E}_s[F_Q]$ (per shot)
circuit depth $L$ (Trotter cycles)
0
2
4
6
8
10
12
0
5
10
15
20
(b) Useful depth grows with $T_2$ (full profile)
saturation depth $L^*$
coherence time $T_2$ (cycles)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0
10
20
30
40
50